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Predictive modeling of burst-mode fs-laser welding process window for glass-solder-metal multilayer materials

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Abstract

To overcome the limitations of time-consuming experiments and inefficient simulations in determining the process window for multi-material, multi-interface joining, this paper presents a predictive modeling methodology for burst-mode femtosecond laser welding of glass-solder-metal multilayer structures. Addressing the intricate interplay of parameters in burst mode, this paper integrates analytical and semi-empirical approaches to rapidly forecast laser energy absorption, thermal accumulation dynamics, and interfacial temperature evolution. The model incorporates multi-interface energy reflection/absorption characteristics and interfacial heat transfer, elucidating key mechanisms: GHz-burst leverages ultrashort (400ps) inter-pulse spacing to boost energy coupling efficiency, while MHz-burst optimizes the balance between plasma shielding

Outline

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and heat build-up. This framework elucidates that burst modes enable simultaneous bonding across both the glass-solder and solder-metal interfaces. Experimental validation demonstrates excellent correlation (deviation <14%) between the model's predicted interface temperatures and observed welding results. Crucially, the model offers high computational efficiency, simulating hundreds of thousands of pulses per parameter set in just seconds, thus offering a high-fidelity, rapid predictive tool for optimizing complex multi-material femtosecond laser welding processes.

Keywords

Femtosecond laser; Burst mode; Predictive model; Process window; Welding

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1. Introduction

The joining of dissimilar materials, particularly glass to metal, is increasingly crucial in modern manufacturing across diverse fields, including microelectronics, photonics, and biomedical devices [1,2]. However, due to the significant disparities in both physical and chemical properties between glass and metal, traditional joining methods such as active brazing [3], matching sealing [4], anodic bonding [5], adhesive bonding [6], among others, struggle to achieve a highly reliable bonding between the two materials. Laser welding, especially with femtosecond lasers, has emerged as a promising technique to overcome these limitations, offering advantages such as localized energy deposition, minimal heat-affected zone, and non-contact processing [7,8].

Femtosecond laser pulses, with their ultrashort duration, enable precise material modification through nonlinear absorption, leading to efficient energy coupling and reduced thermal damage, making them particularly suitable for joining delicate materials

Declaration of competing interest

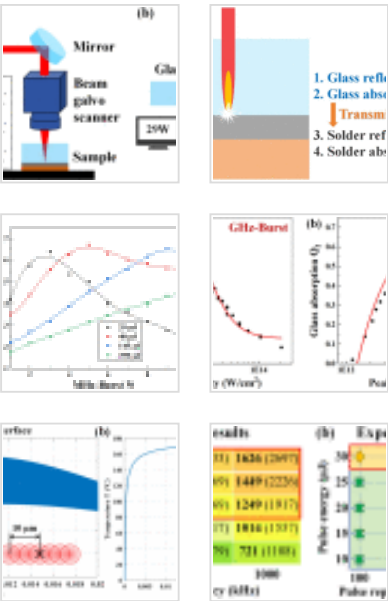
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Figures (11)



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Tables (1)

Table 1. Specific parameter ranges in the...

like glass [9,10]. However, the molten pool created by a femtosecond laser is small, leading to a limited capacity to fill the weld gap. Consequently, a good surface roughness of both glass and metal is necessary to achieve optical contact (with a gap of less than $1/4$ wavelength) [11]. In most opto-mechanical systems, optical components necessitate high positioning accuracy with a specific orientation to fulfill requirements such as converging or reflecting light beams. However, adjusting the position and orientation of the lens creates a significant gap between the lens and the metal base, which greatly increases the difficulty of femtosecond laser welding. In previous research, we suggested the insertion of a paste solder between glass and metal to fill the weld gap, thereby enabling the adjustment of glass position [12]. Subsequently, a burst-mode femtosecond laser is utilized to connect the solder-metal interface and the solder-glass interface in one step [13].

For such a multi-material, multi-interface welding scenario, the primary concern is the welding process window that can achieve simultaneous connections across multiple interfaces. However, the process window for burst-mode femtosecond laser welding is challenging due to the large number of parameters involved, including pulse energy, repetition rate, scanning speed, burst duration, number of pulses per burst, etc. [14,15]. Traditional experimental approaches to determine the optimal process window are time-consuming and costly. Therefore, the vast majority of experimental studies only select 1–2 key parameters within a narrow range to explore their parameter windows and influence patterns, resulting in parameter windows that are clearly local and not universally applicable [[16], [17], [18]]. Analytical models and numerical simulation methods are frequently employed to predict parameter windows and investigate welding mechanisms. Based on theoretical derivations or empirical formulas, analytical models can rapidly compute the free electron density and temperature field for single-pulse or multi-pulse scenarios, providing high computational efficiency and precision [19]. Nevertheless, current research using analytical models is confined to single or similar materials, such as the internal modification of glass or glass-glass welding [20,21]. For more complex glass-metal welding scenarios, there are variations in mechanisms related

to energy absorption and transfer compared to glass-glass welding. Currently, there are few analytical models specifically tailored for glass-metal welding. Numerical simulation methods, such as finite element methods (FEM) and computational fluid dynamics (CFD), can solve temperature, stress, and flow fields under complex boundary conditions and parameters [22,23]. However, due to their computational efficiency constraints, they are generally utilized for calculating single-pulse or a limited number of multi-pulses cases [24]. In scenarios involving thermal accumulation over hundreds of thousands of pulses, the computation time can extend to several hours or even days. Such computational efficiency clearly falls short of the requirements for multi-parameter, large-scale parameter window predictions.

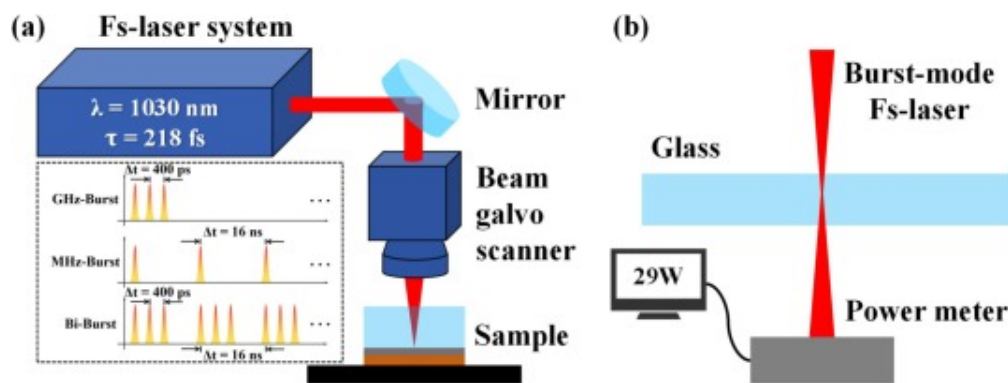
To tackle these challenges, this paper introduces a predictive model for burst-mode femtosecond laser welding of glass-solder-metal multilayer materials. The model is designed to substantially decrease the experimental workload, deepen process understanding, and facilitate efficient prediction of the process window. By integrating analytical models for the swift estimation of the temperature field and semi-empirical models for a detailed analysis of energy absorption and transfer across various interfaces, the paper aims to create a comprehensive predictive tool to optimize the burst-mode femtosecond laser welding process. The predictive model is validated through comparison with experimental results and is further employed to investigate the impact of burst-mode parameters on the welding quality and joint characteristics within glass-solder-metal multilayer structures.

2. Materials and methods

The experimental setup employs a solid-state laser (Carbide-3, Light Conversion) operating at a fundamental wavelength of 1030nm with adjustable pulse durations ranging from $\tau = 218\text{fs}$ to 10ps, as illustrated in Fig. 1(a). For this investigation, the pulse duration is fixed at the minimum value of 218fs. The laser output is directed through a galvanometric scanning system and focused using an F-theta lens to achieve a spot

diameter of $40\mu\text{m}$ on the target surface. Three distinct burst modulation schemes are implemented in this study:

- (1) GHz-burst mode with 400ps inter-pulse intervals (2.5GHz repetition rate);
- (2) MHz-burst mode featuring 16ns inter-pulse intervals (62.5MHz repetition rate);
- (3) Bi-burst mode that incorporated GHz-burst sequences within each MHz-burst cycle.



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Fig. 1. (a) Schematic diagram of burst-mode femtosecond laser processing system for welding glass-solder-metal multilayer material; (b) schematic diagram of the experimental setup for measuring glass absorption.

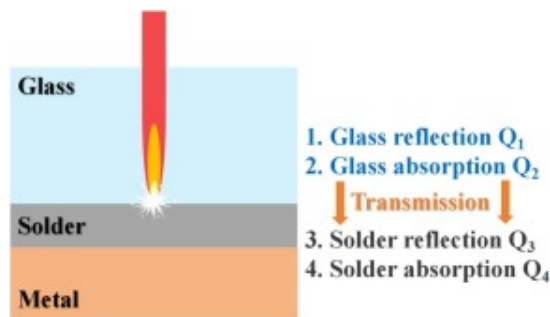
It should be specifically noted that in burst mode, the energy of a single sub-pulse decreases proportionally with the increase in the number of sub-pulses. For example, when the number of sub-pulses is N , compared to the single pulse energy E in non-burst mode, the single sub-pulse energy in burst mode becomes E/N . Here, N refers to the number of sub-pulses for MHz-burst. For GHz-burst, it is P , and for Bi-burst, it is $N*P$.

The experimental materials to be welded consist of quartz glass measuring 20mm×20mm×5mm and copper measuring 20mm×20mm×1 mm. The solder paste is applied between the glass and copper, with a thickness of 100μm. The solder used is a tin-based solder paste (with a melting point of 217°C) that includes flux designed to remove oxides from the surface of the copper. Its metal composition is 99% Sn, 0.3% Ag, and 0.7% Cu. A tensile tester (HANDPI, HDV-5K) is employed to assess the shearing strength of the welded specimens by exerting a tensile force parallel to the weld surface at a speed of 1 mm/min. The shearing force value is determined by averaging at least three measurements to minimize random error. The welding strength is calculated by dividing the shearing force by the total weld area. For a detailed analysis of the welding region, weld seam images and 3D surface profiles are examined using a digital microscope system (KEYENCE, VHX-7000).

3. Process window prediction model

The interaction of burst-mode femtosecond lasers with multi-layer materials composed of glass-solder-metal leads to the absorption, reflection, and transmission of laser energy. Consequently, the input energy of the laser can be categorized into four components, as illustrated in Fig. 2. The femtosecond laser is focused onto the glass-solder interface. During the propagation of the laser, a portion of pulse energy is absorbed by the glass, leading to multiphoton ionization or avalanche ionization, which in turn generates a substantial number of free electrons. When the density of free electrons reaches a critical threshold, it results in significant laser reflection. Consequently, the energy is divided into glass reflection Q_1 , and glass absorption Q_2 . Additionally, the remaining laser energy transmits through the glass and reaches the surface of the solder. The energy reflected by the solder is Q_3 , and the energy absorbed by the solder is Q_4 . Theoretically, there is also a portion of energy reflected by the solder that is reabsorbed by the glass. However, because this reflection occurs within the duration of a single femtosecond pulse, the plasma generated inside the glass at this moment is not significant, and the laser energy density reflected back by the solder is greatly reduced. Therefore, this portion of

reabsorbed energy can be disregarded. It is important to note that the heat and mass transfer behavior at the glass-weld-metal interface involves multi-physics coupling, including molten flow and interfacial phase transitions. Consequently, existing theoretical models find it challenging to accurately describe these dynamic processes. To address this, the current model employs theoretical analytical calculations, experimental data fitting for key parameters, and reasonable assumptions to ensure prediction accuracy. Given that the model utilizes a significant number of experimental fitting formulas for glass absorptivity, it currently shows superior applicability for the quartz glass system. Should researchers wish to apply it to different systems, it is advised to calibrate each parameter through additional experiments.



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Fig. 2. Schematic diagram of laser energy distribution.

3.1. Glass reflection and absorption model

Under the irradiation of femtosecond laser, glass primarily generates initial free electrons via multiphoton ionization. These electrons absorb energy from the laser field until they possess enough to collisionally ionize adjacent atoms, resulting in the production of additional free electrons. This cascade continues until the plasma reaches a critical density, beyond which further laser energy deposition is significantly reduced. Therefore, as the laser intensity increases, the reflection of the laser by the glass plasma also intensifies. Perry et al. have studied the relationship between reflectivity and the peak

power density of the incident laser [25]. For ease of calculation, the data has been fitted to obtain the following relationship:

$$Q_1 = A_0 + A_1 \lg(I_{\text{peak}}) + A_2 (\lg(I_{\text{peak}}))^2 + A_3 (\lg(I_{\text{peak}}))^3 \quad (1)$$

In this equation, Q_1 is the proportion of energy reflected by glass, I_{peak} refers to the peak power density of the laser, $A_0 = -22434.7589$, $A_1 = 4430.6870$, $A_2 = -2.906804$, $A_3 = 6.3598 \times 10^{-2}$. When the peak power density of the laser is below $7.7 \times 10^{12} \text{ (W/cm}^2\text{)}$, the reflectivity is almost zero. The calculation formula for the peak power density of laser is as follows:

$$I_{\text{peak}} = \frac{P_{\text{ave}}}{f \cdot \tau \cdot S} \quad (2)$$

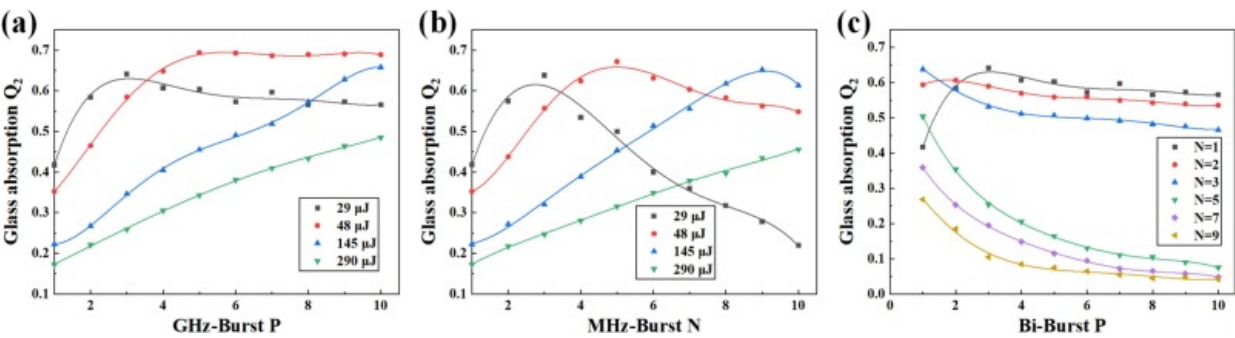
In this equation, P_{ave} is average power, f is pulse repetition frequency, τ is pulse width, S is beam area.

The indirect measurement of laser energy absorption in glass is achieved through the experimental configuration illustrated in Fig. 1(b) [26]. This methodology involves direct quantification of the transmitted laser power detected by the power meter after passing through the glass specimen. Through energy conservation analysis, the absorbed energy component can be calculated by deducting both the transmitted energy and pre-characterized reflected energy from the total incident laser energy. Since altering the number of sub-pulses slightly affects the total energy (power) of the burst. For example, at a set power of 29W, a GHz-burst with 10 sub-pulses generates an actual power of approximately 29.78W, causing a change in the total burst energy. However, since this study has measured the actual output power of the laser under all configurations, the pulse energy is pre-regulated before the experiments to ensure consistent total burst energy across all configurations.

The relationship between glass absorption Q_2 and the number of sub-pulses under three different burst-modes is shown in Fig. 3. The specific parameter ranges involved in the figure are shown in Table 1. Fig. 3(a) demonstrates the correlation between glass

absorption Q_2 and the number of GHz sub-pulses P under varying total burst energies ($29\mu\text{J}$, $48\mu\text{J}$, $145\mu\text{J}$, and $290\mu\text{J}$). Notably, when processing at higher total burst energies, Q_2 exhibits a notable enhancement with increasing sub-pulse counts. Conversely, under lower total burst energy conditions, Q_2 initially rises then stabilizes at an elevated plateau with increasing sub-pulse counts. This phenomenon primarily occurs because when the single-pulse energy is excessively high, the laser intensity becomes extremely strong, and the plasma density within the glass reaches the critical plasma density. At this point, any additional laser input is reflected by the glass plasma, making further laser energy deposition negligible. Consequently, as the number of sub-pulses within the burst increases, the laser intensity is proportionally diminished, resulting in more laser energy being absorbed by the glass rather than reflected. These findings collectively indicate that GHz-burst possess dual functionality: energy-dependent absorption enhancement (at higher total energies) and sustained high-level absorption maintenance (at reduced total energy levels). Fig. 3(b) delineates the relationship between glass absorption Q_2 and MHz sub-pulse counts N under varying total burst energy conditions. While mirroring the absorption enhancement observed in GHz-burst, the MHz-burst exhibits different tendency in low total energy where absorption demonstrates an initial ascent followed by marked attenuation. Mechanistic analysis reveals this disparity originates from critical differences in inter-pulse temporal spacing. For GHz-bursts with 400ps intervals, residual free electrons or self-trapped exciton from preceding pulses persist (relaxation time $< 1\text{ ns}$), enabling subsequent pulses to bypass multiphoton ionization stages and directly initiate avalanche ionization, thereby enhancing energy coupling efficiency [27,28]. Conversely, the 16ns inter-pulse spacing in MHz-bursts exceeds characteristic electron relaxation timescales in quartz glass, forcing each pulse to repetitively undergo complete ionization processes from ground state. Therefore, for MHz-burst, when the number of sub-pulses is too low, it leads to stronger glass plasma reflection; when the number is too high, it results in excessively low peak power density of individual sub-pulses, making the ionization process more difficult. Fig. 3(c) demonstrates the correlation between glass absorption Q_2 and sub-pulse count in Bi-burst mode at a fixed total pulse burst energy of $29\mu\text{J}$. The experimental data reveals two distinct energy

absorption mechanisms based on the number of MHz-burst N. In the low-N regime, the GHz-burst maintains relatively high total energy, enabling efficient plasma generation within the glass substrate. Under these conditions, incremental increases in GHz sub-pulse count show negligible impact on absorption characteristics, as evidenced by the plateau region in the absorption curve. Conversely, in the high-N regime, the progressive addition of GHz sub-pulses induces an proportional decrease in per-pulse energy, falling below the critical threshold for plasma initiation. This energy depletion mechanism results in incomplete photon-plasma interactions, manifesting as a marked reduction in glass absorption efficiency.



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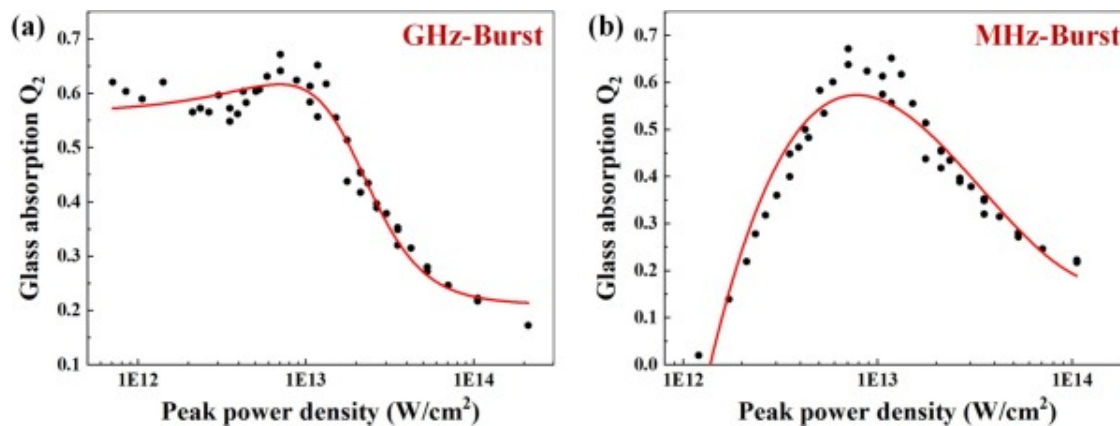
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Fig. 3. Relationship between glass absorption Q_2 and the number of sub-pulses under three different burst-modes. (a) Q_2 versus the number of GHz sub-pulses P under GHz-Burst mode at different total pulse burst energies; (b) Q_2 versus the number of MHz sub-pulses N under MHz-Burst mode at different total pulse burst energies; (c) Q_2 versus the number of MHz sub-pulses N and GHz sub-pulses P under Bi-Burst mode at a fixed total pulse burst energy of 29 μ J.

Table 1. Specific parameter ranges in the glass absorption test.

Experimental series	Laser power	Repetition rage	Scan speed	Focused spot diameter	Key variables	Sample size
Fig. 3(a)	29W	100–1000kHz	1 mm/s	40 μ m	GHz sub-pulse count P	n=40 repeat three times
Fig. 3(b)	29W	100–1000kHz	1 mm/s	40 μ m	MHz sub-pulse count N	n=40 repeat three times
Fig. 3(c)	29W	1000kHz	1 mm/s	40 μ m	Sub-pulse count N and P	n=60 repeat three times

To establish a unified framework for comparative analysis of energy deposition mechanisms, this study introduces peak power density I_{peak} (normalized to individual sub-pulses) as a comprehensive parameter integrating many critical laser variables: pulse energy, repetition rate, pulse duration, sub-pulse count and beam area. [Fig. 4.](#) systematically compares the glass absorption Q_2 under GHz-Burst and MHz-Burst modes as functions of laser peak power density. The red curve in the figure is obtained by fitting the experimental data. The computational formula derived from fitting the experimental data points is as follows:



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Fig. 4. (a) The relationship between glass absorption Q_2 and laser peak power density in GHz-Burst mode; (b) the relationship between glass absorption Q_2 and laser peak power density in MHz-Burst mode.

For GHz-Burst:

$$Q_2 = a + \frac{2b}{\pi} \cdot \frac{\omega}{4(I_{\text{peak}} - c) + \omega^2} \quad (3)$$

In this equation, $a = 0.21087$, $b = 2.28983 \times 10^{13}$, $c = 6.98114 \times 10^{12}$ and $\omega = 3.59474 \times 10^{13}$. The equation loses applicability when the total energy density of GHz-burst approaches or drops below the critical plasma formation threshold. This limitation arises because the nonlinear absorption enhancement mechanism becomes inoperative under sub-threshold conditions where free electron density fails to reach the dielectric breakdown criterion. In glass-metal welding scenarios, the ablation threshold of metal is significantly lower than that of glass. Consequently, metal plasma is more readily generated and expands into the glass, enabling laser energy absorption. Therefore, even when the total energy density of the GHz pulse burst is below the quartz glass ablation threshold ($F_{\text{th}}(\infty) = 1.47 \pm 0.16 \text{ J/cm}^2$) [29] by a certain margin, a high level of energy absorption is still maintained. Experimental results infer that a high plasma intensity can be sustained when the total energy density of the GHz-burst surpasses 0.9 J/cm^2 ($4.1 \times 10^{12} \text{ W/cm}^2$).

For MHz-Burst:

$$Q_2 = B_0 + B_1 \lg(I_{\text{peak}}) + B_2 (\lg(I_{\text{peak}}))^2 + B_3 (\lg(I_{\text{peak}}))^3 \quad (4)$$

In this equation, $B_0 = -924.03353$, $B_1 = 205.80811$, $B_2 = -15.2388$ and $B_3 = 0.37526$. The formula can be used to calculate the energy absorption of glass under MHz-Burst mode and conventional non-burst mode.

In the Bi-Burst mode, the laser energy absorption mechanism in glass exhibits distinct characteristics depending on the total energy density of the GHz-burst, which can be categorized into two typical scenarios:

- (1) When the total energy density of the GHz-burst exceeds the plasma generation threshold, the enhancement or sustaining effects of GHz sub-pulses on energy absorption become predominant. In this regime, the energy absorption can be effectively described by Eq. (3).
- (2) When the total energy density remains below the plasma generation threshold, the suppression effect caused by reduced sub-pulse peak power density dominates the energy absorption process. This regime is more appropriately modeled using Eq. (4).

Based on the experimental findings, the critical threshold energy density for plasma generation in quartz glass-metal welding is predicted to be approximately 0.9 J/cm^2 .

3.2. Solder reflection and absorption model

Based on the theoretical derivations by Fang et al. [30] and Suslova et al. [31], the metal's laser absorptivity demonstrates a monotonic increase with rising laser intensity within the peak power density range of $1 \times 10^{11} \text{ W/cm}^2$ to $1 \times 10^{14} \text{ W/cm}^2$. This established theoretical framework motivated the analysis of experimental data from Vorobyev and Guo's femtosecond laser irradiation experiments on Sn [32]. Numerical fitting of the experimental results reveals the following relationship between the residual thermal effects coefficient and laser fluence:

$$C_{Re} = B_0 + B_1 F + B_2 F^2 \quad (5)$$

In this equation, $B_0 = 0.221959$, $B_1 = 6.2129 \times 10^{-2}$, $B_2 = 3.14 \times 10^{-4}$. F is laser fluence, and C_{Re} is the residual energy coefficient, calculated as $C_{Re} = E_R/E_I$. E_R

stands for the residual thermal energy retained in the sample, while E_I denotes the incident pulse energy.

Furthermore, it can be deduced that the calculation formulas for the solder reflection Q_3 and the solder absorption Q_4 are as follows:

$$Q_3 = (1 - C_{Re}) \cdot (1 - Q_1 - Q_2) \quad (6)$$

$$Q_4 = 1 - Q_1 - Q_2 - Q_3 \quad (7)$$

3.3. Interface thermal accumulative temperature calculation model

The thermal accumulation induced by pulse bursts is governed by two critical parameters: sub-pulse repetition rate and pulse number. These factors jointly determine the magnitude of temperature rise and cooling rate through their coupling with thermal relaxation dynamics. Based on the Toy model proposed by Kerse et al. [33], which assumes each pulse burst generates an instantaneous temperature rise ΔT increment proportional to its pulse energy E_p . Then the sub-pulse energy is E_p/N , and the resulting temperature rise is $\Delta T/N$. Here, N refers to the number of sub-pulses. For GHz-burst, it is P , and for Bi-burst, it is $N \cdot P$. The cumulative temperature evolution for a single pulse burst can be expressed as [34]:

$$T_n(t) = \sum_{n=1}^{N_{sub}} \frac{\Delta T \cdot H(t - (n-1)\tau_R)}{\sqrt{1 + \frac{t - (n-1)\tau_R}{\tau_0}}} \quad (8)$$

In this equation, ΔT is a temperature rise immediately following pulse irradiation, H is the heaviside function, τ_R is the time intervals between sub-pulses, and τ_0 is the thermal relaxation time. N_{sub} is the number of sub-pulses.

Although the time to calculate a single pulse burst is short, the thermal accumulation temperature field that needs to be obtained typically requires the calculation of hundreds of thousands of pulse bursts, which can result in a very long overall computation time. Therefore, in order to improve computational efficiency, the pulse burst is packaged into a

single pulse, and an influence coefficient A is introduced to differentiate the temperature rise caused by the pulse burst from that caused by a single pulse.

$$A = \frac{T_n(t_{p-p})}{T_1(t_{p-p})} \quad (9)$$

where T_n is the temperature rise caused by the pulse burst, T_1 is the temperature rise caused by a single pulse in non-burst mode, and t_{p-p} is the time interval between pulse bursts ($t_{p-p} = 1/f$), and f is the pulse repetition frequency. The temperature rise at t_{p-p} is the temperature rise calculated before the arrival of the next pulse burst.

Heat transfer between glass and solder primarily occurs through heat conduction and convection. Due to the complexity of these processes, which are influenced by factors such as the temporal and spatial temperature distributions of both materials, as well as the temperature dependence of thermal conductivity and convective heat transfer coefficients, a heat transfer enhancement factor, K , is introduced to account for the impact of glass-solder heat transfer on the temperature field of solder and metal.

$$K = \frac{Q_2 + Q_4}{2Q_4} \quad (10)$$

Ultimately, temperature calculations at the upper and lower surfaces of the solder layer allow for the determination of whether effective bonding has been achieved at both the glass-solder and solder-metal interfaces. This study assumes that the temperature at the upper surface of the solder layer corresponds to the glass-solder interface temperature, while the lower surface temperature of the solder layer corresponds to the solder-metal interfacial temperature. Following Bauer et al.'s thermal accumulation model [35,36], a computational model for the solder-side temperature field under femtosecond laser multi-pulse conditions is modified and developed:

$$T(x, y, z, t) = \frac{2A \cdot K \cdot E_p \cdot Q_4}{\pi \rho c_p \sqrt{\pi \kappa t} (8\kappa t + \omega_0^2)} \times \exp\left[\frac{(x-x_c)^2 + (y-y_c)^2}{4\kappa t} \left(\frac{\omega_0^2}{8\kappa t + \omega_0^2} - 1\right)\right] \exp\left[\frac{-z^2}{4\kappa t}\right] \quad (11)$$

In this equation, t is the time after the energy pulse, E_p is the laser input pulse energy, c_p is the material average heat capacity, ρ is the mass density and κ is the thermal

diffusivity. $(\mathbf{x} - \mathbf{x}_c)$, $(\mathbf{y} - \mathbf{y}_c)$ and $\mathbf{z}$ represent the distance from the center of the laser spot $(\mathbf{x}_c, \mathbf{y}_c, 0)$, and ω_0 is the beam radius at 1/e of the maximum intensity. For multi-pulse irradiation, the time distance between pulses is t_{p-p} . The temperature distribution after N pulses is thus given by the following:

$$T(\mathbf{x}, \mathbf{y}, \mathbf{z}, t) = \sum_{n=1}^N T_{\mathbf{x}_{cn}, \mathbf{y}_{cn}}^{\text{SinglePulse}}(\mathbf{x}, \mathbf{y}, \mathbf{z}, t + n t_{p-p}) \quad (12)$$

3.4. Model modification guide

When the material system is changed or the laser's inherent parameters such as wavelength and pulse width are altered, the fundamental relationships among glass reflection Q_1 , glass absorption Q_2 , solder reflection Q_3 , and solder absorption Q_4 remain unchanged. However, their specific values and the corresponding energy distribution ratios will be affected. To enhance the applicability of this model and to facilitate its correction by researchers, the model correction procedure is summarized as follows:

- 1) The relationship between glass reflectivity and the incident laser's peak power density is influenced by the type of glass and the inherent laser characteristics. Therefore, the glass reflectivity needs to be re-measured and fitted to obtain the corrected Eq. (1).
- 2) Changing the glass type and inherent laser characteristics will significantly alter key parameters such as the glass ablation threshold. Consequently, glass absorption Q_2 will be considerably affected. This not only changes the magnitude of absorptivity at different laser intensities but also alters the laser intensity corresponding to the maximum absorptivity. However, the principles governing the interaction of Single pulse, MHz-burst, and GHz-burst with the glass material remain the same: Single pulse will generate strong plasma reflection due to its excessively high laser intensity; MHz-burst achieves better absorption effects due to weakened plasma reflection; GHz-burst, due to the enhanced thermal ionization effect, will exhibit a high energy absorption rate even at lower energy densities. Therefore, during the

correction process, re-fitting Eqs. (3), (4) can be completed by supplementing with only a small number of experiments in the low, medium, and high laser intensity ranges.

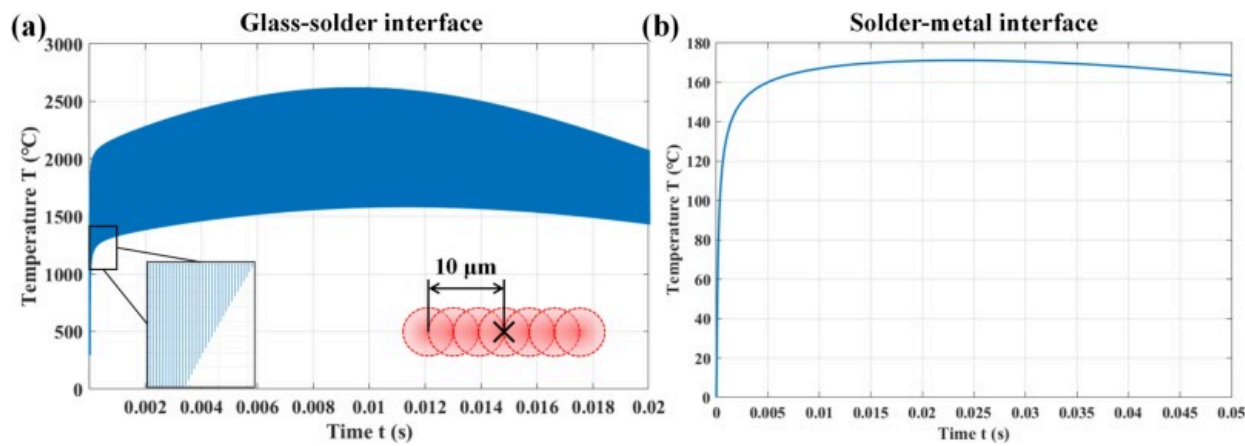
- 3) As the laser absorptivity of different metals varies significantly, it is necessary to re-conduct residual energy measurement experiments based on the specific metal and laser used. This is done to obtain the relationship curve of energy absorption as a function of laser intensity, thereby allowing for the correction of Eq. (5).
- 4) Based on the fundamental principle of energy conservation, Eqs. (6), (7) can be used without modification.
- 5) Eqs. (8), (9) are used to calculate the impact of varying sub-pulse time intervals on thermal accumulation for different burst patterns. If it is necessary to change “the time intervals between sub-pulses τ_R ” and “the number of sub-pulses N_{sub} ,” one only needs to modify these two parameters and then perform an analytical calculation using Eq. (8) to obtain the influence coefficient A.
- 6) Eq. (10) represents a reasonable simplification of the heat transfer process between the glass and the solder and can be used without modification.
- 7) Eqs. (11), (12) are analytical formulas with clear physical significance, used for calculating the temperature field during the laser multi-pulse heating process. In the correction process, one only needs to change the specific values of parameters such as “average heat capacity c_p ,” “mass density ρ ,” and “thermal diffusivity κ ” to accurately calculate the temperature field data for the target material under specific working conditions.

4. Results and discussion

4.1. Calculation results under non-burst mode

For the multi-interface temperature field simulation of glass-solder-metal systems, this study employs representative process parameters (single-pulse energy: $29\mu\text{J}$, repetition rate: 1 MHz, scanning speed: 1 mm/s, non-burst mode: $N=1$, $P=1$) for numerical analysis. To accurately characterize the thermal accumulation effects during laser scanning, a linear scanning mode is adopted, with a characteristic point located $10\mu\text{m}$ from the scanning start position selected as the temperature monitoring site (spatial positioning illustrated in the inset of Fig. 5(a)). Simulation results reveal: As shown in Fig. 5(a), the temperature evolution curve at the glass-solder interface exhibits two characteristic temperature boundaries: the upper boundary corresponds to the transient temperature (2596°C) immediately after laser pulse irradiation, while the lower boundary reflects the temperature (1592°C) before the next pulse arrives. In this study, these two temperatures are defined as the “peak temperature” and the “steady temperature”, respectively. First and foremost, it is important to note that the research background encompasses a complex issue with three layers of materials and two interfaces. In determining whether effective bonding has been established, evaluations must be conducted independently for both the glass-solder interface and the solder-metal interface. Effective bonding can only be considered achieved when the temperatures at both interfaces surpass their respective melting points. For the glass-solder interface, the bonding mechanism is predicated on a fusion welding process induced by ultrafast laser irradiation. Therefore, effective bonding can only occur if both the glass and the solder are melted. Consequently, this study posits that effective bonding at this interface is achieved only when the temperature exceeds the softening point of quartz glass. Regarding the metal-solder interface, the bonding mechanism is based on a brazing process driven by elemental diffusion. Thus, effective bonding at this interface can be realized as long as the solder melts. Notably, the peak temperature at this interface significantly exceeds the softening temperature of quartz glass (1700°C), confirming effective bonding under these parameters. Further analysis of the solder-metal interface (Fig. 5(b)) reveals a smooth single-peak temperature profile with a peak value of 171°C , owing to the $100\mu\text{m}$ spatial separation from the laser interaction interface. This temperature remains substantially below the melt temperature

of the solder (217°C), indicating insufficient thermal activation for achieving effective bonding between the solder layer and metal substrate.



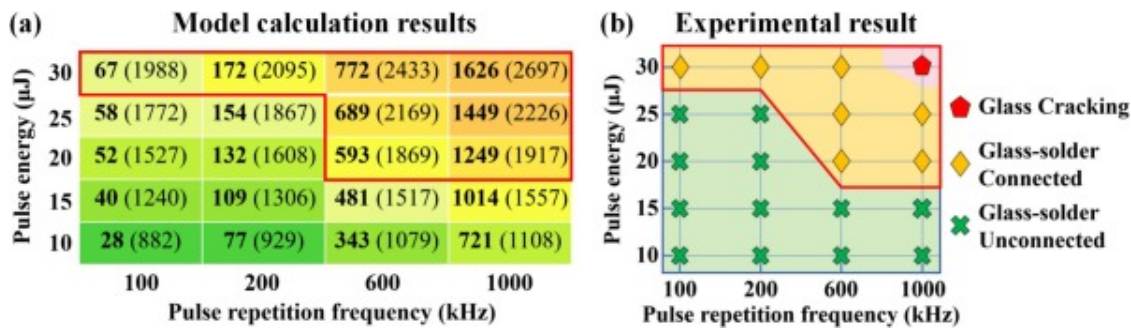
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Fig. 5. (a) Temperature variation curve of the glass-solder interface over time; (b) temperature variation curve of the solder-metal interface over time.

Further developing this research, systematic model simulations and experimental investigations are conducted across an extensive parameter range with single pulse energy (10–30 μJ) and pulse repetition frequency (100–1000 kHz) as independent variables. Notably, the parameter combination of 30 μJ and 1000 kHz approaches the maximum output power limit (approximately 30 W) of the laser system employed in the experiments. Fig. 6(a) presents the model-calculated temperature results at the glass-solder interface. The bold numerals indicate steady temperatures, while parenthetical values represent transient peak temperatures. Critical calculation points where peak temperatures exceed the softening threshold of quartz glass (1700°C) are highlighted with red borders, facilitating direct comparison with experimental observations (glass-solder connected). The incorporated red border line inherently accounts for a 10% deviation margin between computational predictions and experimental results at the

1700°C reference point. For example, there are discrepancies between the calculated and experimental results under parameters 25μJ, 100kHz and 25μJ, 200kHz.



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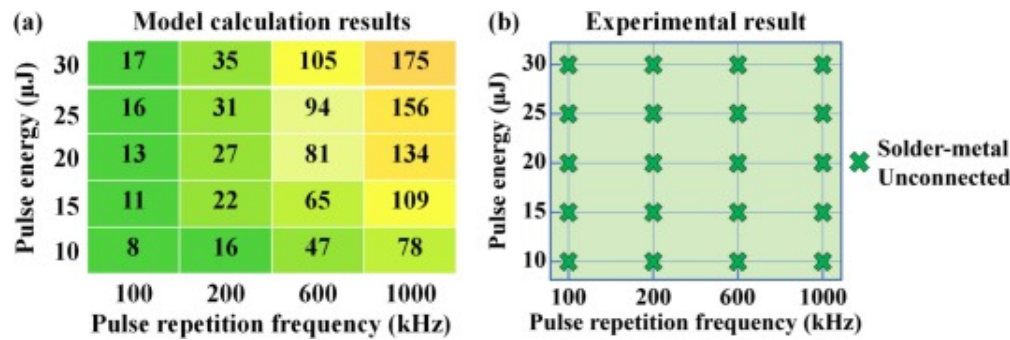
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Fig. 6. Variable parameter simulation and experimental results of the glass-solder interface under non-burst mode. (a) Interface temperature results obtained from model calculations. The bold numerals represent the stable temperature, and the numerals in parentheses represent the peak temperature. Calculation points with peak temperatures exceeding the softening temperature of quartz glass (1700°C) are outlined with red borders for ease of comparison with experimental results. This red line automatically includes the 10% deviation of the calculation results from the experimental results; (b) connection status of the glass-solder interface obtained from experiments. (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

Theoretical analysis reveals two distinct thermal mechanisms: elevated single-pulse energies predominantly enhance peak temperatures (critical for localized melting), while increased repetition frequencies significantly improve thermal accumulation efficiency, thereby raising steady temperatures (essential for large-area fusion). This frequency-dependent thermal accumulation behavior aligns well with established findings in prior literature [37]. However, under most non-burst mode parameters, stabilized temperatures remained subcritical (<1700°C), indicating insufficient thermal accumulation to achieve

comprehensive melting of quartz glass which is a critical limitation for effective elimination of micro-cracks and ablation defects. Fig. 6(b) demonstrates the experimentally obtained interface connection states under various parameter combinations. Comparative analysis reveals remarkable consistency between model-predicted interface temperatures and actual experimental outcomes. The developed model demonstrates reliable predictive accuracy within a 10% error margin for determining interface melting status and connection effectiveness.

Fig. 7(a) presents the model-calculated thermal results at the solder-metal interface. The results demonstrate a positive correlation between thermal accumulation and laser power (product of energy and frequency), with higher repetition frequencies showing superior thermal accumulation efficiency under equivalent power conditions. Within the investigated parameter range, interface temperatures consistently remained below the solder's melting point (217°C), indicating complete bonding failure with zero measurable strength at the solder-metal interface. This theoretical prediction finds strong empirical verification in Fig. 7(b), where experimental observations confirm absent interfacial connections across all parameter sets. This combined model-experimental evidence reveals a critical limitation of non-burst mode processing: insufficient thermal accumulation prevents concurrent multi-material bonding at both glass-solder and solder-metal interfaces. The fundamental thermal deficit observed at the solder-metal interface particularly highlights the non-burst mode's inability to sustain simultaneous dual-interface melting requirements, even when partial glass-solder bonding might occur under specific conditions.



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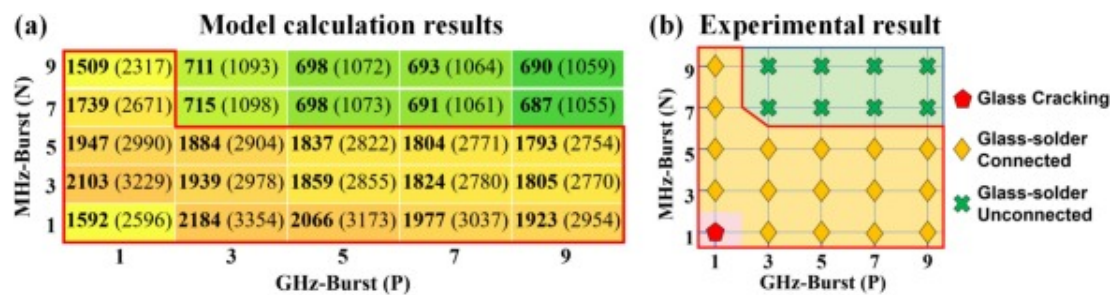
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Fig. 7. Variable parameter simulation and experimental results of the solder-metal interface under non-burst mode. (a) Interface temperature results obtained from model calculations; (b) connection status of the solder-metal interface obtained from experiments.

4.2. Calculation results under burst mode

In the study of investigating burst mode laser welding, the total burst energy is maintained at 29 μJ with a repetition rate of 1000 kHz. Systematic investigations are conducted across three burst modes (MHz-Burst, GHz-Burst, and Bi-Burst) using selected parameter ranges for MHz sub-pulses ($N=1-9$) and GHz sub-pulses ($P=1-9$). Fig. 8(a) presents the computed temperature at the glass-solder interface. The bold numerals indicate steady temperatures, while parenthetical values denote peak temperatures. Calculation points exceeding the quartz glass softening threshold (1700°C) are highlighted with red borders for experimental correlation. The simulation reveals that controlled augmentation of either MHz (N) or GHz (P) sub-pulse counts within defined ranges significantly elevates both peak and steady interface temperatures compared to those in non-burst mode, enabling effective glass melting across extended parameter domains. At lower N values ($N \leq 5$), incremental P demonstrate minimal thermal influence. With higher N value, each additional GHz sub-pulse induces substantial

temperature reduction. As evidenced by the experimental results in Fig. 8(b), the developed model demonstrates remarkable predictive accuracy in burst-mode processing. Among them, when the number of MHz sub-pulses $N=7$ or 9, the energy density of the laser is already at a relatively low level. When the number of GHz sub-pulses P increases from 1 to 3, it directly causes the energy density of each sub-pulse to drop to 1/3 of the original. This significant reduction in energy density makes the laser intensity unable to reach the ionizing threshold of the glass, so the glass almost no longer absorbs laser energy. Under these circumstances, the absorption of laser energy mainly relies on the metal side. Due to the significant difference in absorption characteristics between metal and glass, a very pronounced temperature gradient change occurs.



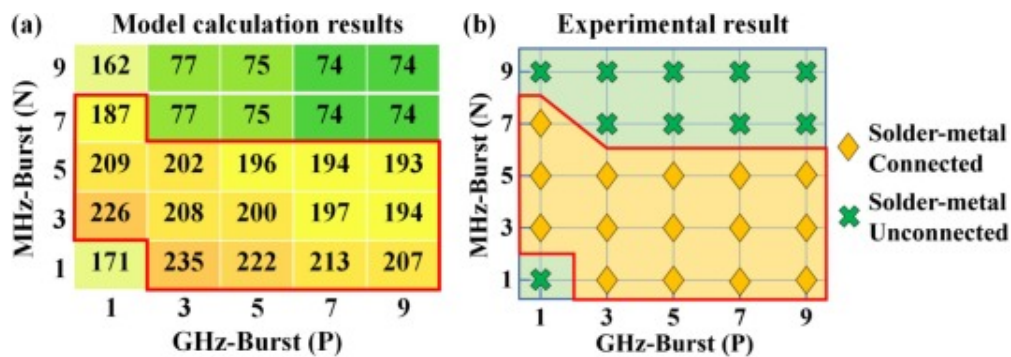
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Fig. 8. Variable parameter simulation and experimental results of the glass-solder interface under burst mode. (a) Interface temperature results obtained from model calculations. The bold numerals represent the stable temperature, and the numerals in parentheses represent the peak temperature. Calculation points with peak temperatures exceeding the softening temperature of quartz glass (1700°C) are outlined with red borders for ease of comparison with experimental results. (b) Connection status of the glass-solder interface obtained from experiments. (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

Fig. 9(a) illustrates the thermal results at the solder-metal interface obtained through computational model. Critical regions where calculated temperatures exceed the solder's

melting point (217°C) are demarcated with red boundaries, incorporating a 14% tolerance margin to facilitate comparative analysis with experimental data. Comparative thermal analysis reveals that optimized burst parameters significantly enhance interfacial thermal accumulation compared to conventional non-burst mode. Computational model demonstrate successful bonding at the solder-metal interface across all three burst modes under specified parameter configurations ($N=1-5$). Integrated with previously analyzed temperature characteristics at the glass-solder interface, this study establishes a critical finding: The burst mode enables simultaneous bonding of the dual interfaces of glass-solder-metal multimaterials with an intermediate layer thickness of $100\mu\text{m}$ across an extensive parameter window, a technical achievement unattainable in non-burst mode. Experimental validation in Fig. 9(b) confirms strong model-experiment correlation within 14% error margin, conclusively validating the predictive accuracy of this computational model under complex multi-physics coupling conditions.



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Fig. 9. Variable parameter simulation and experimental results of the solder-metal interface under burst mode. (a) Interface temperature results obtained from model calculations. Calculation points exceeding the melting temperature of solder (217°C) are outlined with red borders for ease of comparison with experimental results. This red line automatically includes the 14% deviation of the calculation results from the experimental results; (b) connection status of the solder-metal interface obtained from experiments.

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To investigate the interfacial bonding mechanisms and fracture behavior under burst mode laser welding, this study systematically analyzes welding strength and fracture morphology. Fig. 10 presents shearing strength from parametric burst-mode experiments, with at least three tests conducted for each parameter set. The analysis revealed critical insights: In non-burst mode ($N=1, P=1$), intense plasma generation within the glass substrate reflected over 30% of laser energy while creating severe internal ablation defects, resulting in insufficient bonding strength at both glass-solder and solder-metal interfaces. By contrast, optimized burst parameters demonstrated significant improvements: GHz-Burst mode achieved peak strength of 12.4MPa at $N=1/P=5$; MHz-Burst mode attained 8.3MPa at $N=5/P=1$; Bi-Burst mode reached 11.2MPa at $N=3/P=3$. Calculation model indicated these optimal parameters generated comparable interfacial temperatures: (2066°C, 222°C) for GHz-Burst, (1947°C, 209°C) for MHz-Burst, and (1939°C, 208°C) for Bi-Burst. This thermal consistency suggests maintaining ~2000°C at the glass-solder interface enables stable molten pool formation while avoiding plasma-induced defects. For solder-metal interfaces, achieving the melting temperature of solder sufficiently activates interfacial diffusion, resulting in reliable metallurgical bonding.

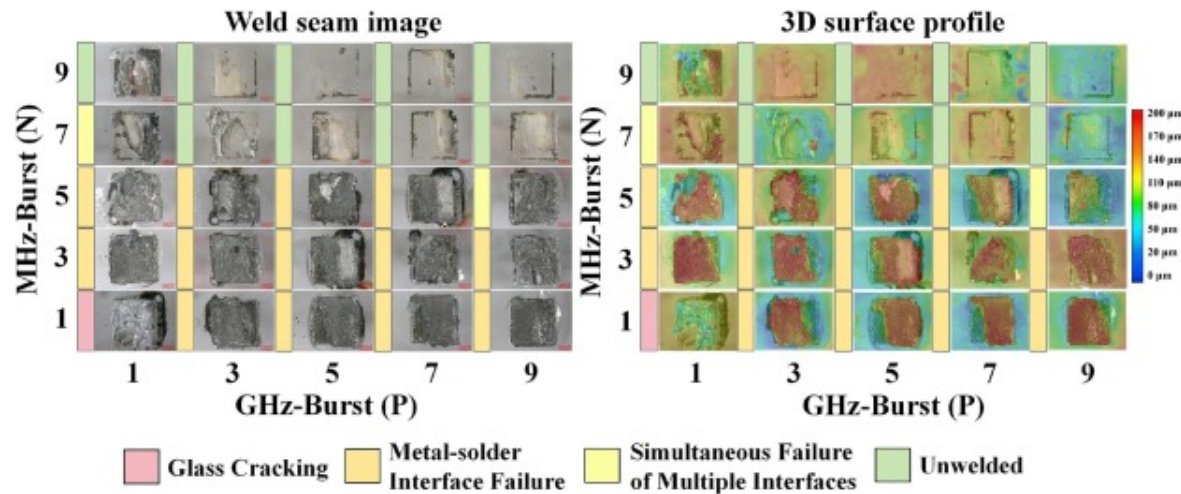
Shearing strength (MPa)					
MHz-Burst (N)	0	0	0	0	0
	2.8 ± 0.3	0	0	0	0
	8.3 ± 1.2	9.5 ± 1.1	6.8 ± 1.2	7.1 ± 0.7	5.3 ± 1.6
	6.1 ± 0.8	11.2 ± 1.8	7.8 ± 0.8	6.9 ± 0.9	8.2 ± 0.7
	0	8.5 ± 1.5	12.4 ± 2.2	10.8 ± 1.7	9.0 ± 2.0
	1	3	5	7	9
GHz-Burst (P)					

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Fig. 10. The shearing strength of the variable parameter experiment under burst mode.

The fracture morphology analysis of glass-solder-metal joints presents significant challenges in interface identification due to the complex multi-material interactions. Conventional microscopic observation alone proves insufficient for precise fracture interface determination. To address this limitation, this investigation employed a comprehensive approach combining cross-sectional morphology characterization with 3D surface profile of the glass-side fracture surfaces, as illustrated in Fig. 11. In non-burst mode ($N=1$, $P=1$), zero shearing strength and severe glass fragmentation indicates inadequate bonding at both glass-solder and solder-metal interfaces, suggesting complete interface failure. For the majority of specimens ($N=1-5$, $P=1-9$), a consistent $100\mu\text{m}$ elevation of the welded zone relative to the glass substrate confirms predominant solder-metal interface failure. This failure mode hierarchy arises from the superior bonding strength at the glass-solder interface compared to its solder-metal counterpart. Specimens processed under ($N=7$, $P=1$) and ($N=5$, $P=9$) parameters exhibit simultaneous interfacial failure, as evidenced by mixed fracture characteristics. The computational model reveals interface steady temperatures of 1739°C and 1793°C , respectively, approaching the quartz glass softening point (approximately 1700°C). Therefore, the bonding strength at the glass-solder interface has also decreased to the same level as the solder-metal interface. For the samples under other parameters, since the temperatures at both interfaces are significantly lower than the melting temperature, no effective connection is formed at either interface. These experimental observations substantiate the predictive accuracy of the predictive model, particularly regarding interface bonding-temperature relationships. The methodology successfully correlates process parameters with failure mechanisms through quantitative interface analysis, providing critical insights for joint optimization in glass-metal welding applications.



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Fig. 11. Weld seam images and 3D surface profiles from burst mode variable parameter experiments.

5. Conclusion

This study successfully develops and validates a predictive model for determining the process window of burst-mode femtosecond laser welding for glass-solder-metal multilayer structures. The research addresses the challenge of achieving simultaneous, reliable bonding across both the glass-solder and solder-metal interfaces. The key findings are as follows:

- (1) Burst modes significantly enhance energy coupling and thermal accumulation. GHz-burst mode leverages ultrashort (400ps) inter-pulse spacing to boost energy coupling efficiency, while MHz-burst mode optimizes the balance between plasma shielding and heat build-up. Appropriately selected burst parameters successfully raise temperatures

above the respective melting/softening points at both interfaces simultaneously, enabling reliable joining.

- (2) Model-calculated interface temperatures correlates well with experimentally observed bonding status at both interfaces across a wide range of parameters, with prediction deviation for melting thresholds within approximately 10% for the glass-solder interface and 14% for the solder-metal interface. Crucially, the model boasts high computational efficiency, simulating hundreds of thousands of pulses per parameter set within seconds.

Given that current models partially rely on experimental data fitting for key parameters and include rational assumptions, some formulas lack physical significance and demonstrate poor transferability to other materials or configurations. Consequently, the following elements will be integrated into subsequent research: (1) Deriving theoretical correlation formulas for certain parameters through the combination of finite element simulations; (2) Expanding the experimental dataset to train prediction models based on material properties and laser parameters, thereby progressively decreasing reliance on empirical data and further expanding the model's applicability scope.

CRediT authorship contribution statement

Haodong Ren: Writing – original draft, Visualization, Validation, Methodology, Investigation, Formal analysis, Data curation, Conceptualization. **Hong Shen:** Writing – review & editing, Supervision, Methodology, Investigation, Funding acquisition, Conceptualization.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Data availability

The data that support the findings of this study are available from the corresponding author upon reasonable request.

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